

Association of parental education with adolescents' COVID-19 vaccine uptake: A nationwide register-based study in Finland

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ABSTRACT

Background: Evidence regarding the link between parental education and adolescents' COVID-19 vaccination status mostly originates from non-representative survey studies, and the results are mixed. Although some register-based studies have shown a positive association with higher parental education and adolescent COVID-19 vaccine uptake, the factors behind these differences remain largely unexplored.

Methods: We used total population-level Finnish register data on 12–17-year-olds ($N = 360,281$) linked with their parents and household members. Logistic regression models were employed to analyse the association between parental education and the likelihood of receiving the first COVID-19 vaccine dose by the end of 2022. The Karlson–Holm–Breen (KHB) method was used to quantify the extent to which parental age, family structure, household income, migration background, previous COVID-19 infection, presence of risk group household members, and parental COVID-19 vaccination status confounded or mediated the association.

Results: Adolescents with at least one parent with upper tertiary education had higher COVID-19 vaccine uptake (92 %) than those whose parents had only basic education (58 %), with an odds ratio [OR] of 8.60 (95 % CI 8.20–9.02) in the baseline model. Together the adjustments confounded or mediated over 80 % of the association, with parental COVID-19 vaccine uptake and household income being the strongest explanatory factors. For example, adjusting for parental COVID-19 vaccination status attenuated the OR to 3.85 (95 % CI 3.61–4.12). In the fully adjusted model, the OR remained at 1.55 (95 % CI 1.45–1.67). The results were highly similar when using household income as the socioeconomic status measure.

Conclusion: Parental education was a strong predictor of adolescents' COVID-19 vaccine uptake in Finland. Parental vaccination status emerged as an important mediator of the association, possibly indicating general vaccine hesitancy. Other factors behind the educational differences included household income, migration background, and parental age.

1. Introduction

Individuals with low socioeconomic status (SES) have been disproportionately affected by the COVID-19 pandemic in terms of incidence and severity in Finland [1] and globally [2–5]. Previous studies also indicate that adults with low SES have been less likely to receive COVID-19 vaccination, with education level being one of the most frequently reported predictors [6–8]. However, previous results on educational differences in children's and adolescents' COVID-19 vaccinations are mixed and mostly based on non-representative samples.

Three systematic reviews have reported contradictory results on the association of parental education with the willingness to vaccinate children against COVID-19 [9–11]. The survey studies that have assessed actual COVID-19 vaccine uptake alongside willingness have generally found higher vaccination coverage among children of higher-educated parents [12,13], but opposite results have also been reported [14]. The survey studies, however, are often limited by low response rates and the samples being unrepresentative of the general population.

Current population-representative register-based evidence comes from Norway and Sweden. Two Norwegian studies have found higher

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parental education and household income to be positively associated with 12–17-year-old adolescents' COVID-19 vaccine uptake [15,16]. A recent Swedish study found an association between lower household education and lower COVID-19 vaccination coverage among adolescents in Stockholm aged 12–17 years [17]. In these, the COVID-19 vaccination statuses of parents and household members were identified as partial drivers of the educational differences in adolescents' vaccine uptake [15,17]. However, which other family characteristics influence this association and to what extent remains unclear.

The reasons for the educational differences in adolescents' COVID-19 vaccination status may relate to various family characteristics such as economic and social resources, level of health literacy, and trust in health authorities. Besides parental vaccination status, which may indicate general vaccine hesitancy, we address several other factors that are more common among families with low parental education and may either confound or mediate the association between parental education and adolescent COVID-19 vaccine uptake. Household income captures material aspects of SES [18], which may mediate the educational differences. For example, financial constraints and the stress associated with earning a living could lower the likelihood of obtaining a vaccination among low-educated families. Furthermore, economic and social resources are likely to be lower in single-parent families compared to two-parent families. Several review studies have identified young parental age to be linked with lower willingness to vaccinate children against COVID-19 [9–11]. Migration background may confound educational differences through factors related to language barriers, health literacy, and structural discrimination [19–21].

Previous COVID-19 infections have been linked to a lower likelihood to obtain COVID-19 vaccination [22]. The larger number of COVID-19 infections among low SES groups in Finland [1] supports the consideration of COVID-19 infection history behind educational differences in vaccination coverage. Several comorbidities and health conditions along with older age increase the risk for severe COVID-19, and people in these groups were the first in Finland to be recommended for COVID-19 vaccination [23]. Given that COVID-19 vaccination has been shown to reduce infectiousness [24], it is reasonable to assume that having high-risk household members may have influenced the decision to vaccinate others in the household. Families of low SES have greater likelihood for poor health [25], which could potentially confound the association between parental education and adolescent vaccine uptake.

Using nationwide register data on socioeconomic measures and COVID-19 vaccination, we assess how parental education was associated with COVID-19 vaccine uptake among 12–17-year-old adolescents in Finland by the end of 2022. Drawing from existing literature, we address the role of various family characteristics including parental age, family structure, household income, migration background, previous COVID-19 infections, risk group status for severe COVID-19, and parental COVID-19 vaccine uptake behind the educational differences. Our study aims to improve understanding of the socioeconomic differences in adolescent COVID-19 vaccine uptake.

2. Methods

2.1. Study setting

In Finland, COVID-19 vaccinations began in December 2020, with adolescents aged 16 to 17 years vaccinated at the same pace as the adult population, starting with those at a high risk for severe COVID-19 disease. For those aged 12 to 15 years, vaccinations were recommended for at-risk groups in June 2021 [26], and the government decree allowing COVID-19 vaccinations for the entire age group entered into force in August 2021 [27]. In some municipalities, children's vaccinations took place in schools [28]. Vaccinations were free of cost and widely available.

2.2. Data sources

We used Finnish administrative register data with personal identification codes enabling linkages across various registers as well as between household members and parents and children. From Statistics Finland we derived end-of-year information for 2020 on education, age, sex, hospital district, urbanicity, family structure, and migration background, and for 2019 on household members and household income, according to data availability. For identifying COVID-19 vaccination status, we combined information from the Register of Primary Health Care Visits (AvoHilmo) maintained by the Finnish Institute for Health and Welfare (THL) and the Patient Data Repository (Kanta) maintained by the Social Insurance Institution of Finland. We derived the conditions associated with an increased risk for severe COVID-19 illness from the special reimbursement register of the Social Insurance Institution of Finland. COVID-19 diagnoses were identified from the National Infectious Diseases Register of THL.

2.3. Study population

We studied the likelihood of COVID-19 vaccine uptake among adolescents from 1 January 2021 to 31 December 2022. Based on the most recent available information, the study population was composed according to the following criteria. We included all individuals who turned 12–17 years during 2021 (born between 2004 and 2009) and were permanently residing in Finland at the end of 2020, excluding those who died or emigrated by the end of 2021 ($N = 486$). All adolescents were linked to their biological parents, and those without information on either biological parent were excluded ($N = 9708$). The final study population consisted of $N = 360,281$ individuals. For further analyses, the adolescents were divided into two age groups: 12–15 (born between 2006 and 2009) and 16–17 years (born between 2004 and 2005) due to the later start of vaccinations among the younger age group.

2.4. Outcome

The outcome was defined as having received at least one dose of COVID-19 vaccination by the end of 2022. COVID-19 vaccinations were identified primarily by anatomical therapeutic chemical (ATC) codes J07BN01, J07BN02, J07BN03, J07BN04, and J07BX03 from the Register of Primary Health Care Visits. We did not have access to vaccination batch numbers, which could have been used as a quality assurance of the ATC records. To minimize possible incorrect records, we supplemented our data with COVID-19 vaccination records from the Patient Data Repository maintained by the Social Insurance Institution of Finland. The final number of administered COVID-19 vaccinations in our dataset closely aligns with those publicly reported in Finland by THL [29].

2.5. Covariates

Parental education was measured as the highest completed education of the parents by the end of 2020 and categorized as 1) basic, 2) secondary, 3) lower tertiary, and 4) upper tertiary education.

Adolescent's sex was measured as binary, and age was categorized into one-year age groups. Hospital district and urbanicity were based on each adolescent's place of residence at the end of 2020. Hospital district ($n = 21$) was included to adjust for regional differences in the spread of the pandemic. Urbanicity was categorized as 1) urban area, 2) peri-urban area (including local centres in rural areas), and 3) rural area.

We measured parental age at birth in accordance with the commonly used demographic definitions of young and advanced parental age. Maternal and paternal ages were categorized separately as 1) under 25 years, 2) 25–29 years, 3) 30–34 years, and 4) 35 years or older. If maternal information was missing, paternal age was used, and if paternal information was missing, maternal age was used.

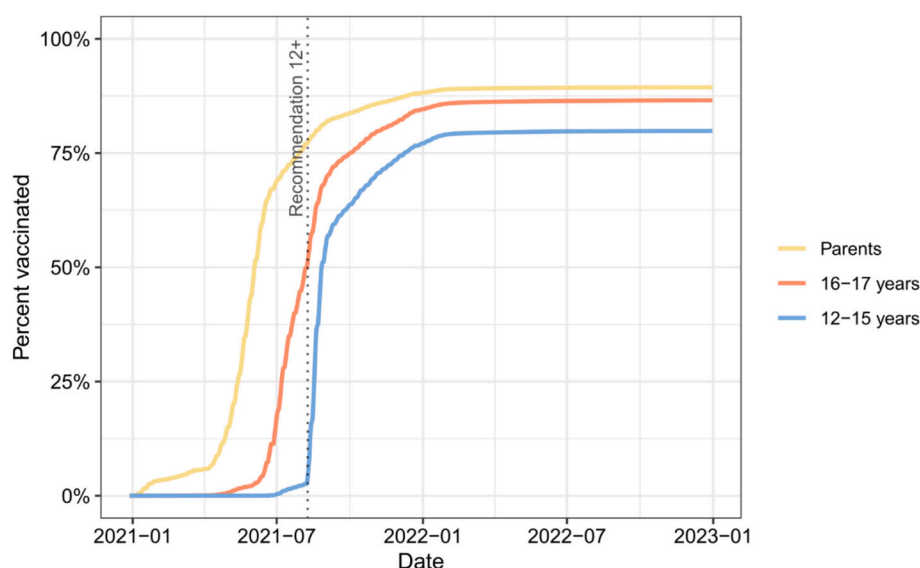


Fig. 1. COVID-19 vaccination coverage (having received at least one dose) from 1 January 2021 to 31 December 2022 among parents and their adolescent children aged 12–15 years and 16–17 years. The dotted line shows the date when COVID-19 vaccinations were made available and recommended for all individuals aged 12+ years in Finland.

Each adolescent's family structure at the end of 2020 was categorized as 1) living with one parent, 2) living with two parents, and 3) not living with a parent (e.g. living alone or with others).

Annual household income, including all taxable income of all household members in 2019, was divided by the number of consumption units using the Organisation for Economic Co-operation and Development's (OECD) modified equivalence scale and categorized into quintiles.

Migration background (no/yes) was defined as adolescent who themselves, both parents, or the only known parent were born outside Finland and whose native language was not Finnish or Swedish.

Adolescent's and household members' previous COVID-19 infection (no/yes) was defined as having a laboratory-confirmed COVID-19 case (ICD-10 code U07.1) between February 1, 2020, and February 21, 2021, based on the latest available data.

Adolescent's and household members' risk group status (no/yes) was defined as having health conditions associated with an increased risk for severe COVID-19 illness. The health conditions were selected based on the risk group definitions by THL during the study period [23]. The conditions were identified from information from the Social Insurance Institution regarding the right to special reimbursement for medicinal expenses related to specific diagnosed chronic conditions [30]. These conditions included cancer, kidney failure, chronic lung disease, diabetes (types 1 and 2), chronic heart disease (heart failure, hypertension, coronary heart disease), and psychotic disorders. Cases with a reimbursement right in effect at the end of 2020 were included. For household members, everyone aged 70 or older at the end of 2020 were also considered to have risk group status.

Parental COVID-19 vaccination status was defined as having received the first dose of COVID-19 vaccination by the end of 2022. It was categorized as 1) neither parent (or the only known parent) vaccinated, 2) one parent vaccinated while the other is not, and 3) both parents (or the only known parent) vaccinated.

2.6. Statistical analyses

We used logistic regression to model the association between parental education and adolescents' COVID-19 vaccination status, adjusted for various covariates at a time. To account for the likely correlation between siblings, we clustered standard errors at the household level.

To quantify the extent to which each of our adjustments confounds or mediates the association between parental education and adolescents' COVID-19 vaccination status, we used the reformulated Karlson–Holm–Breen (KHB) method [31]. The KHB method addresses the rescaling bias associated with comparing non-linear models by decomposing the total effect into direct and indirect effects. The direct effect represents the effect of X on Y that is not mediated by other variables, while an indirect effect refers to the portion mediated by other factors. In this method, linear predictor V is first obtained from the full logistic model regressing Y on X and all other predictors. Predictor V is then used as a dependent variable in ordinary least squares (OLS) regression, first without and then with the mediators [31]. The attenuation percentage is calculated as the indirect effect divided by the total effect. In our study, this represents the fraction of the association between parental education and adolescent COVID-19 vaccination status that is confounded or mediated by the variables added in each model.

2.7. Analysis strategy

In Model 1, we adjusted for adolescent's age, sex, hospital district, and urbanicity. Subsequent models were built upon Model 1 by adjusting for each set of covariates that could potentially confound or mediate the association between parental education and adolescent vaccination status in a separate model. In Model 2, we adjusted for parental age. In Model 3, we adjusted for adolescent's family structure. In Model 4, we adjusted for household income. In Model 5, we adjusted for adolescents' migration background. In Model 6, we adjusted for previous COVID-19 infection of the adolescent and previous COVID-19 infection in the household. In Model 7, we adjusted for adolescent's or other household members' high risk for severe COVID-19 illness due to underlying health conditions. In Model 8, we adjusted for the parental COVID-19 vaccination status. Finally, in Model 9, we simultaneously adjusted for all the variables mentioned above.

We also ran similar models as above but using household income quintiles in place of parental education to investigate whether the associations differ when using another common measurement for SES, which captures different dimensions of it by assessing material circumstances [18].

As an additional analysis, we ran all the models separately for 12–15- and 16–17-year-olds, because of their different timing in receiving COVID-19 vaccinations. It is also reasonable to assume that parents

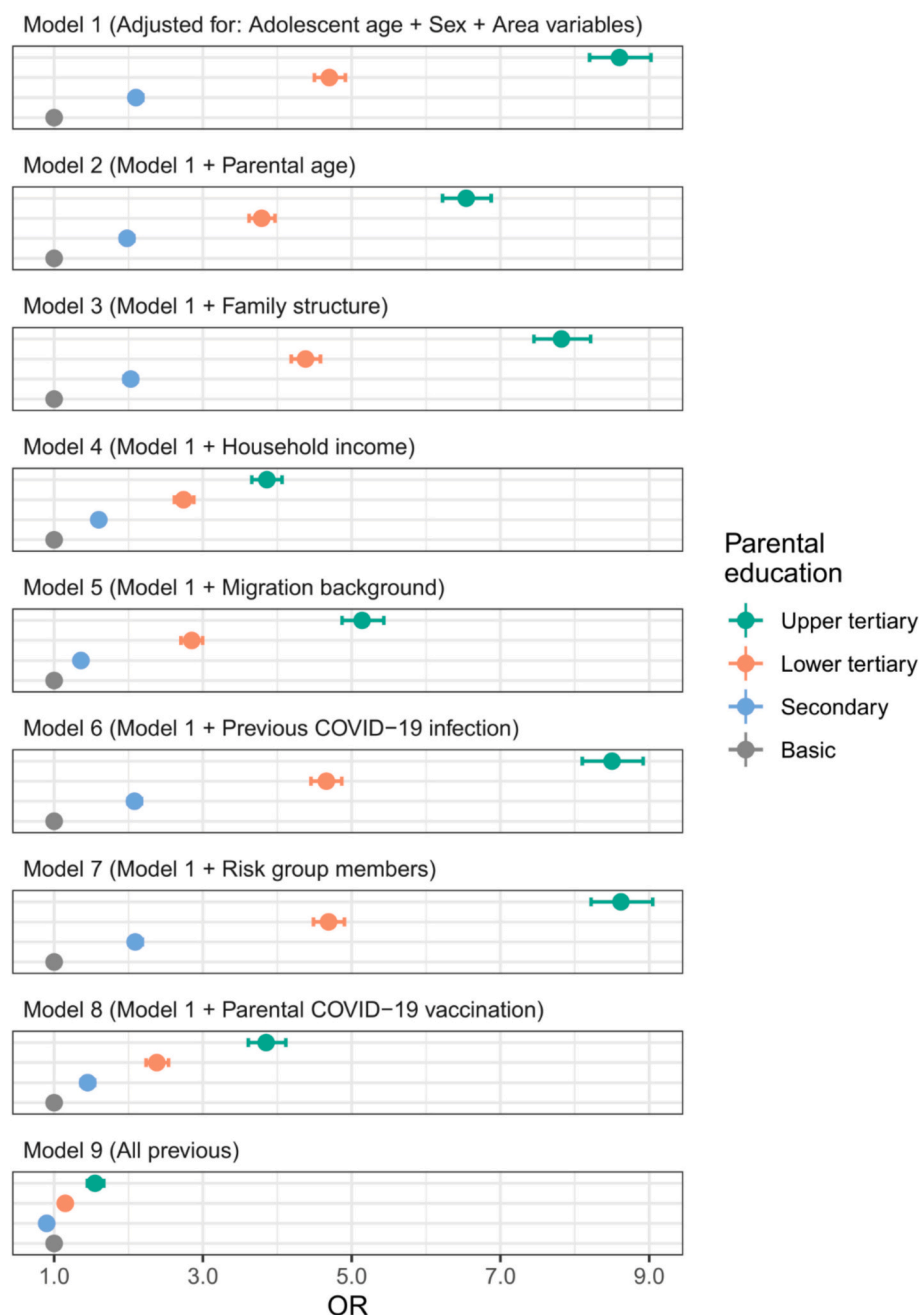


Fig. 2. Odds ratios of having received at least one dose of the COVID-19 vaccine among adolescents aged 12–17 years by parental education, 1 January 2021 to 31 December 2022. Standard errors clustered at the household level.

presumably have greater decision-making authority when it comes to vaccinating younger adolescents. How the various family characteristics influence the association between SES and vaccinations may therefore be different in these age groups.

KHB models were performed using Stata 16.1. All other models were performed with R version 4.3.1.

3. Results

By the end of 2022, 80 % of 12–15-year-olds and 87 % of 16–17-year-old adolescents had received their first dose of the COVID-19 vaccine (Fig. 1, Supplementary Table 1). Those with higher-educated parents had higher vaccination rates compared to those with lower-educated parents. For example, 58 % of adolescents aged 12–17 years whose parents only had basic education received the vaccination compared to

92 % of those whose parents' highest education level was upper tertiary education. The vaccination status of adolescents also varied considerably by other variables. The distribution by all family characteristics is shown in Supplementary Table 1.

In the baseline model adjusted for age, sex, hospital district, and urbanicity, the odds of having received at least one dose of COVID-19 vaccination were nearly nine times higher among those adolescents whose parent had upper tertiary education (OR 8.60, 95 % CI 8.20–9.02) compared to those whose parents had only basic education (Model 1, Fig. 2, Supplementary Table 2). The odds ratios from the further adjusted models 2–9 are presented in Fig. 2 and Supplementary Table 2, and the proportion of the association that was attenuated by each adjustment derived from equivalent KHB models are presented in Table 1 and Supplementary Table 9.

Of the adjusted variables, parental COVID-19 vaccination status

Table 1

Attenuation percentages compared to the baseline model derived from the KHB models for having received at least one dose of the COVID-19 vaccine among adolescent aged 12–17 years by parental education, 1 January 2021 to 31 December 2022.

	Parental age	Family structure	Household income	Migration background	Previous COVID-19 infections	Risk group members	Parental COVID-19 vaccination	All previous
Parental education	% (95 % CI)	% (95 % CI)	% (95 % CI)	% (95 % CI)	% (95 % CI)	% (95 % CI)	% (95 % CI)	% (95 % CI)
Basic	Ref.	Ref.	Ref.	Ref.	Ref.	Ref.	Ref.	Ref.
Secondary	9.0 (7.9–10.1)	4.2 (3.5–4.9)	35.1 (32.5–37.8)	60.1 (55.4–64.7)	1.2 (0.9–1.6)	0.5 (0.3–0.8)	62.7 (58.6–66.8)	110.0 (103.3–116.6)
Lower tertiary	14.6 (13.7–15.5)	4.3 (3.8–4.7)	35.4 (33.8–37.0)	36.0 (34.1–38.0)	0.7 (0.5–0.9)	0.4 (0.2–0.5)	54.5 (52.4–56.5)	93.0 (89.8–96.3)
Upper tertiary	13.4 (12.7–14.2)	4.2 (3.8–4.6)	37.5 (36.1–39.0)	27.7 (26.4–29.1)	0.6 (0.5–0.8)	0.1 (–0.0–0.2)	46.6 (45.1–48.1)	83.1 (80.7–85.5)

Notes: The baseline model adjusted for age, sex, hospital district, and urbanicity. The attenuation percentages were calculated as the indirect effect divided by the total effect following the Karlson–Holm–Breen (KHB) method [31]. The 95 % confidence intervals (CI) were calculated using bootstrap with 1000 replicates. Standard errors were clustered at the household level. Coefficients are reported in Supplementary Table 9.

(Model 8), household income (Model 4), and migration background (Model 5) had the strongest attenuating effects on the association between parental education and adolescent COVID-19 vaccine uptake. For example, compared to the reference group of adolescents whose parents have received basic education, adjustment for parental COVID-19 vaccine uptake attenuated 47 % of the difference in the upper tertiary educated group and 63 % of the difference in the secondary educated group (Table 1). Parental age (Model 2) and family structure (Model 3) had milder attenuating effects.

A previous COVID-19 infection in the adolescent or in their household (Model 6) and having risk group status for severe COVID-19 disease or having high-risk household members (Model 7) had little effect on the educational gradient, probably at least partly due to their relative rareness in the study population. Together, the adjusted variables confounded or mediated over 80 % of the vaccination differences by parental education (Table 1). In the fully adjusted Model 9, the odds of having received a COVID-19 vaccine remained 1.55 times higher among those adolescents whose parents had upper tertiary education (OR 1.55, 95 % CI 1.45–1.67) compared to those whose parents had only basic education, while differences between other groups were modest.

The results from models using household income as the main exposure variable are shown in Fig. 3 and Supplementary Table 3. In the baseline model (Model 1), the differences according to household income quintile were highly similar to the educational differences reported above, with an OR of 8.39 (95 % CI 8.05–8.74) in the highest household income quintile compared to the lowest. However, in the fully adjusted model (Model 9), the OR remained higher at 2.52 (95 % CI 2.40–2.66). Similarly to the education models, parental COVID-19 vaccine uptake, parental education, and migration background had the strongest attenuating effects on the association, the attenuation percentages shown in Supplementary Table 4.

In an additional analysis, we assessed the associations separately among 12–15- and 16–17-year-olds (Supplementary Tables 5–8). Generally, the differences according to education and household income were more pronounced in the older age group.

4. Discussion

Using high-quality total population data, we showed that higher parental education was associated with higher COVID-19 vaccine uptake among adolescents aged 12–17 years in Finland. Adolescents with at least one parent with an upper tertiary education had almost nine times higher odds of receiving the COVID-19 vaccination compared to those whose parents only had basic education. The association was strongly mediated by parental COVID-19 vaccination status and household income and confounded by migration background. The findings were highly similar when using household income as the measurement of SES instead of parental education.

Our finding of the association between higher parental education and adolescent COVID-19 vaccination status are in line with findings from two Norwegian register studies and a study in Stockholm County [15–17]. Other evidence on the subject is largely survey-based and assesses parental willingness instead of actual vaccine uptake. These studies have reported inconsistent results, which may relate to selected study populations, different research designs, and the timing and area of the study in relation to the pandemic stage and vaccine rollout.

Only a few studies have analysed which factors may confound or mediate the educational differences in adolescent COVID-19 vaccination status. In one Norwegian study, the educational differences were somewhat smaller between the adolescents of vaccinated parents than between all adolescents in the study population [15], suggesting that parental vaccination may mediate these differences. In the Swedish study, including the vaccination status of household members slightly reduced the educational differences, but they simultaneously adjusted for multiple other factors [17]. However, to our knowledge no studies have separately assessed the role of several family characteristics such as migration background, household income, COVID-19 infections, parental age, and family structure behind the educational differences in adolescent COVID-19 vaccination status. Instead of including these factors simultaneously in a single model, our strategy allowed specifically estimating the strength and direction of their effects on the association.

We identified parental COVID-19 vaccination status as the strongest mediator behind the educational differences in adolescent vaccination status, followed by household income. The mediating effect of parental COVID-19 vaccination status may reflect concerns regarding the novelty of the vaccine and the possible side effects, which previously have been reported as major reasons for parents' hesitancy towards children's COVID-19 vaccinations [9,32,33]. Household income was an important mediator even though COVID-19 vaccinations were free and widely available in Finland. Moreover, financial constraints are unlikely to explain the social gradients in vaccination since the differences were particularly notable between the highest educated and highest income groups and others. However, other resources affecting the decision to vaccinate (e.g. knowledge of the vaccines, time, transportation) may be lesser in families with lower income. Socioeconomic differences in vaccine uptake might also depend upon different levels of agency in preventive health care or selective incentives for vaccination such as ability to dine in restaurants or travel abroad during the pandemic.

A substantial amount of the educational differences was confounded by migration background. Parental age also had a smaller attenuating effect. The influence of migration background may be driven by language barriers, health literacy, racism, and structural discrimination, which are often experienced by individuals with migration background and low SES [19–21]. Additionally, cultural and religious beliefs may play a role in vaccination decisions among migrant populations [34].



Fig. 3. Odds ratios of having received at least one dose of the COVID-19 vaccine among adolescents aged 12–17 years by household income, 1 January 2021 to 31 December 2022. Standard errors clustered at the household level.

The attenuating effect of parental age aligns with previous studies showing lower vaccination rates among children of young parents [9–11,35]. For example, both younger maternal age and lower education level were associated with an increased risk of the child not receiving the influenza vaccination in Finland during the 2009 pandemic influenza [35]. In contrast, the larger proportion of single-parent families among the less educated had only a small effect on the association between parental education and adolescent vaccination. Number of children in the household may also influence the time-related or logistic challenges families face in accessing vaccination and could be considered as a further adjustment in future studies.

In our results, the association between parental education and

adolescent COVID-19 vaccine uptake was somewhat more pronounced among the older adolescents (16–17-year-olds) compared to the younger group (12–15-year-olds). This is in contrast to the Norwegian study, where the association was more pronounced among the younger adolescents (12–15-year-olds) [15]. The contrasting results may be related to the differences in vaccination practices between the countries. In Norway, no general recommendations were given to vaccinate under-16-year olds [15], whereas the recommendation in Finland covered everyone aged 12 years and older [28]. In addition, parental consent was required in Norway for COVID-19 vaccination administration to under-16-year-olds [15]. Finland has no specific age limit, as each child's right to decide on their own vaccination is assessed on a case-by-

case basis by the vaccinator. Due to the varying recommendations and vaccination practices, studying socioeconomic differences across countries is important to effectively target interventions aimed at increasing vaccination rates.

While parental COVID-19 vaccine uptake, household income or parental education, migration background, and parental age all substantially attenuated the strength of the association, parental education and household income remained associated with adolescent COVID-19 vaccination status in the fully adjusted models. Especially adolescents of parents with upper tertiary education and those with the highest household incomes were more likely to be vaccinated than the others were. The factors behind the SES differences may relate to varying cultural and social factors that we were unable to directly capture. For example, previous research suggest that people with low SES have lower trust in public health authorities [36] and a lower level of health literacy [37]. Another possible factor that we were unable to investigate further relates to how vaccinations were arranged in schools in certain municipalities. For example, as children of less-educated parents tend to have more school absences [38], they may have been more likely to miss the COVID-19 vaccinations conducted in schools. All in all, our results highlight the importance of building trust and confidence towards public health authorities among parents—particularly those with lower education—in the post-pandemic era.

The main strength of our study was the use of accurate total population register data, which avoids the non-response and recall biases of survey studies. This is important because much of the existing evidence relies on survey data. Vaccination questionnaires may be prone to social desirability bias, particularly in settings where vaccinations are officially recommended or required. Another strength of our study was the ability to link individuals not only to their biological parents but also to household members. This allowed us to study household-level factors that may have an important role in vaccination decisions related to infectious disease prevention.

Our study had some limitations related to data availability. Information of household members was from the end of 2019, which may not fully reflect the household composition right before the vaccinations. However, household size remained the same for 85 % of the individuals in the study population from 2019 to 2020, suggesting that the change in household composition was relatively small. In addition, as our data on COVID-19 infections ended in February 2021, we did not capture infections during the rest of 2021 when the majority of adolescents received their first dose of the COVID-19 vaccine. Another limitation of our study was that with register data we cannot directly capture various social and cultural factors such as safety concerns, vaccination beliefs, and trust in public health authorities, which may influence the educational disparities seen in our results.

5. Conclusion

In Finland, nearly all (92 %) adolescents of the highest-educated parents received the COVID-19 vaccination, while just over half (58 %) of adolescents with the least-educated parents did. Our results imply that parental COVID-19 vaccination status, household income, and migration background were the key factors behind the association between parental education and adolescent vaccination. Since lower socioeconomic status is associated with a higher risk for infections and severe disease, addressing vaccine hesitancy among individuals with low education would be crucial for reducing health inequalities in future pandemics.

CRediT authorship contribution statement

Sanni Variskallio: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Heta Moustgaard:** Writing – review & editing, Writing – original draft, Supervision, Methodology,

Conceptualization. **Hanna Remes:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Conceptualization. **Pekka Martikainen:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

Ethical approval

This study is based on secondary data collected for administrative and statistical purposes. We have obtained permission to access these data from Statistics Finland (permission #TK-53-1490-18) and Findata Health and Social Data Permit Authority (permission #THL/2180/14.02.00/2020) after consideration by the ethical boards of these statistical authorities. The study complies with the national legal framework for accessing anonymous personal data for scientific research carried out in public interest. The legal basis is stated in the Finnish Personal Data Act (523/1999), Act on Secondary use of Social and Healthcare data (552/2019), Finnish Statistics Act (280/2004), and the EU General Data Protection Regulation (GDPR). The GDPR permits processing this type of data for research without using the GDPR consent (Art. 9 of the GDPR).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.vaccine.2025.127615>.

Data availability

The authors do not have permission to share data.

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